Aahil Afraz

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EDUCATION

Drexel University | Philadelphia, PA | GPA: 3.14 **Expected Graduation: June 2028**

B.S. in Software Engineering, Minor in Finance

Relevant Courses: Systems Architecture, Data Structures & Algorithms, Software Project Management, Database Management

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, SQL, C, C++, Bash, Flask, Swift, REST APIs, GraphQL, React

AI & Agents: LangChain, LangGraph, MCP, A2A protocol, RAG, Prompt & Agentic Engineering, LLM Evaluation, Langfuse

Cloud, Data & DevOps: AWS (Batch, Aurora MySQL), Azure, Kubernetes, Docker, CI/CD, GitHub Actions, Linux, PostgreSQL

Certifications: Azure Fundamentals (**AZ-900**), Azure AI Fundamentals (**AI-900**), Security, Compliance & Identity (**SC-900**)

PROFESSIONAL EXPERIENCE

Comcast | Philadelphia, PA **March – September 2026**

Software Engineering Co-op, DevSecOps (Received Return Offer)

- Developed an **LLM-powered** security analytics platform covering **9,000+** applications, unifying **5+** dashboards into one view
- Presented the platform concept to **VPs and executive leadership** while operating in an **AI-native pod format** (distinct from traditional Agile sprints), leading to its integration into the primary developer portal serving **30,000+ engineers company-wide**
- Built a **LangChain AI agent** with **A2A** protocol & custom **MCP** toolset of **5+ tools** for real-time retrieval of live security data
- Designed a **GraphQL** schema and **AWS Batch** pipeline to unify **4 security sources** into **Aurora** MySQL materialized views
- Optimized agent pipeline to **sub-3**-second responses via **Langfuse** tracing and deployed to internal **Kubernetes** within **2 months**

Zaro Health | Philadelphia, PA **August 2026 – Present**

Software Engineer (Startup Environment)

- Shipped features for Zaro Health's **iOS App Store** release and launched within **3 months** alongside a **10-person founding team**
- Customized an **open-source tool** (Postiz) into an AI content generation pipeline with **Higgsfield MCP** for social media outreach, helping land a partnership with influencer **Dr. Rena Malik** (~**3M** YouTube subscribers) and producing content for her platforms
- Implemented a **standardized** design system (adopted team-wide) that keeps AI context & output consistent to company branding

LearnFlow-AI | Philadelphia, PA **June – September 2025**

Founding Engineer

- Founded and launched a production AI learning platform to **150+ beta users**, built with **React**, **Electron**, and an **Express.js** backend
- Built an **Agentic RAG pipeline** that let users query their lecture recordings & slides using **OpenAI Whisper** and **Cloud Vision OCR**

PROJECTS

NASA Space Apps Challenge 2025 | AirCast | (Team Relentless) (GitHub-Link) **October 2025**

- Awarded **Global Nominee: Top 1,290 of 11,500+ Teams Worldwide**: built an AI-powered air quality forecasting platform

PayPal AI Hackathon | AgentGuard |(GitHub-Link) **October 2026**

- Built an **MCP security proxy** that intercepts **AI-agent payment** actions via the **PayPal Agent Toolkit**, verifying refunds with a **LangGraph** risk pipeline (Claude agent + rules) and human-in-the-loop approval for high-risk refunds in the **PayPal sandbox**
- Developed an **adversarial evaluation harness** that cut fraudulent payouts from **$809 to $0** at **96% detection precision**

Linux NASDAQ ITCH Order Book |(GitHub-Link) **September 2026**

- Built a **C++20** limit order book rebuilding from real NASDAQ ITCH 5.0 data (**268M+** messages) at **12.7M msgs/sec, p99 417 ns**
- Verified against a reference book across **263M+** top-of-book checks with 0 mismatches; **165** GoogleTest tests in Docker + GitHub CI
- Cut post-warm-up heap allocations **80M → 3.6K** with a preallocated order pool and open-addressing index, profiled with **perf**